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Lesson Notes

MODULE 8 | LESSON 1

BACKTESTING WITHOUT OVERFITTING: DEFLATING SHARPE AND OVERCOMING SELECTION BIAS

Reading Time	30 minutes
Prior Knowledge	Sharpe ratio, Validation, Model optimization
Keywords	Deflated Sharpe

In Module 7, we discussed how to robustify a model: We focused on manipulating the covariance matrix or the related correlation matrix, in order to improve the output from mean-variance optimization. This led us to de Prado's Hierarchical Risk Parity (HRP) as an even more robust portfolio optimization procedure, which integrated some of the important techniques we had learned about in other contexts, for example, risk parity and clustering. In the readings about HRP, as well as in the readings about other portfolio construction models, we saw comparisons of models based on their Sharpe ratio. We have also discussed in this course how statistics such as Sharpe can be misleading. Recall from Module 6 the discussion of p-hacking and other forms of bias. The title of Lesson 1, Module 6 referred to the possibility of "anomalous profits." In this lesson, we learn how to overcome this bias in how we interpret others' research as well as how we perform and share our own research.

1. What is Backtesting?

Backtesting is a way to evaluate the performance of a predictive model or strategy using historical data. If you are backtesting a model, you might estimate the accuracy of your predictions or the goodness of fit between your predictions and actual observations using a measurement such as mean squared error (MSE), mean absolute percent error (MAPE), mean absolute deviation (MAD), among others. Similarly, if you are backtesting a strategy, you might use any number of ratios or statistics. The most popular is probably the Sharpe Ratio because as a quotient, it captures two objectives: maximizing excess return (return above the risk-free rate) and minimizing risk (where risk is understood as the standard deviation of returns); that is, the Sharpe ratio (SR) measures risk-adjusted excess return. Some other possible metrics discussed as recently as the last lesson include returns (just returns), variance, value-at-risk, expected shortfall, average drawdown, maximum drawdown, deviation from a benchmark—again, among many others. If you are backtesting a strategy, you are running the strategy using historical data and finding the value of one

of these measures to determine if the strategy is in some sense valid (e.g., profitable and/or not too risky).

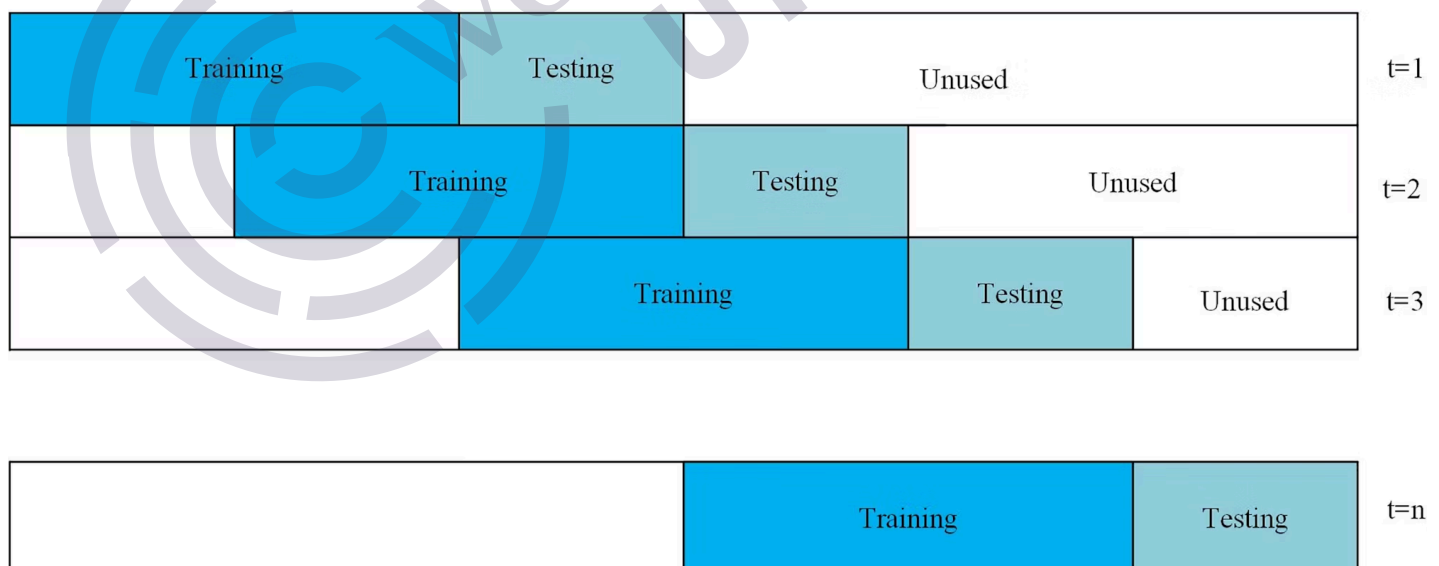
1.1 Popular Approaches to Backtesting

Key to any testing is to use out-of-sample data to validate the model or the strategy that relies on a model. Testing your strategy using the same data that you trained it on will almost certainly overestimate the performance to be expected when your strategy trades live in the market. In this sense, a good backtest reveals the extent of over- or underfitting in your model. The two most popular approaches to backtesting with which you are no doubt already familiar are walk forward and cross validation.

“Walk forward” can be used to describe any testing with rolling optimization or periodic re-optimization (Lv 7). You might also have seen it referred to as “multi-period out-of-sample optimization” or “rolling horizon out-of-sample analysis” (Gambeta 17). The main idea is that you test the model on a rolling or walking forward basis on multiple, out-of-sample, succeeding periods. Since it re-optimizes frequently, it “provides an efficient way to continuously adapt a trading model to ongoing changes in market conditions” and measures the robustness of the model’s ability to handle these changes (Lv 7). Here is a concise and precise description followed by a visualization:

To compare different policies following the “rolling-horizon” procedure, we choose a window T over which to perform the estimation such that $T < L$, where L is the total number of samples in the dataset. Say we use a window of $T = 120$ data points (ten years of monthly data). Then, we compute the different portfolio policies for the data points in the first window. We repeat this for the next period, by including the data for the new dates and dropping the earlier period data, and continue until the end of the dataset. By the end of this process, we have $L - T$ portfolio weight vectors for each strategy; that is, w_t^k for $t = T, \dots, L - 1$ and for each strategy k . (DeMiguel 568-569)

Figure 1: Rolling Out-of-Sample or Walk-Forward Backtesting

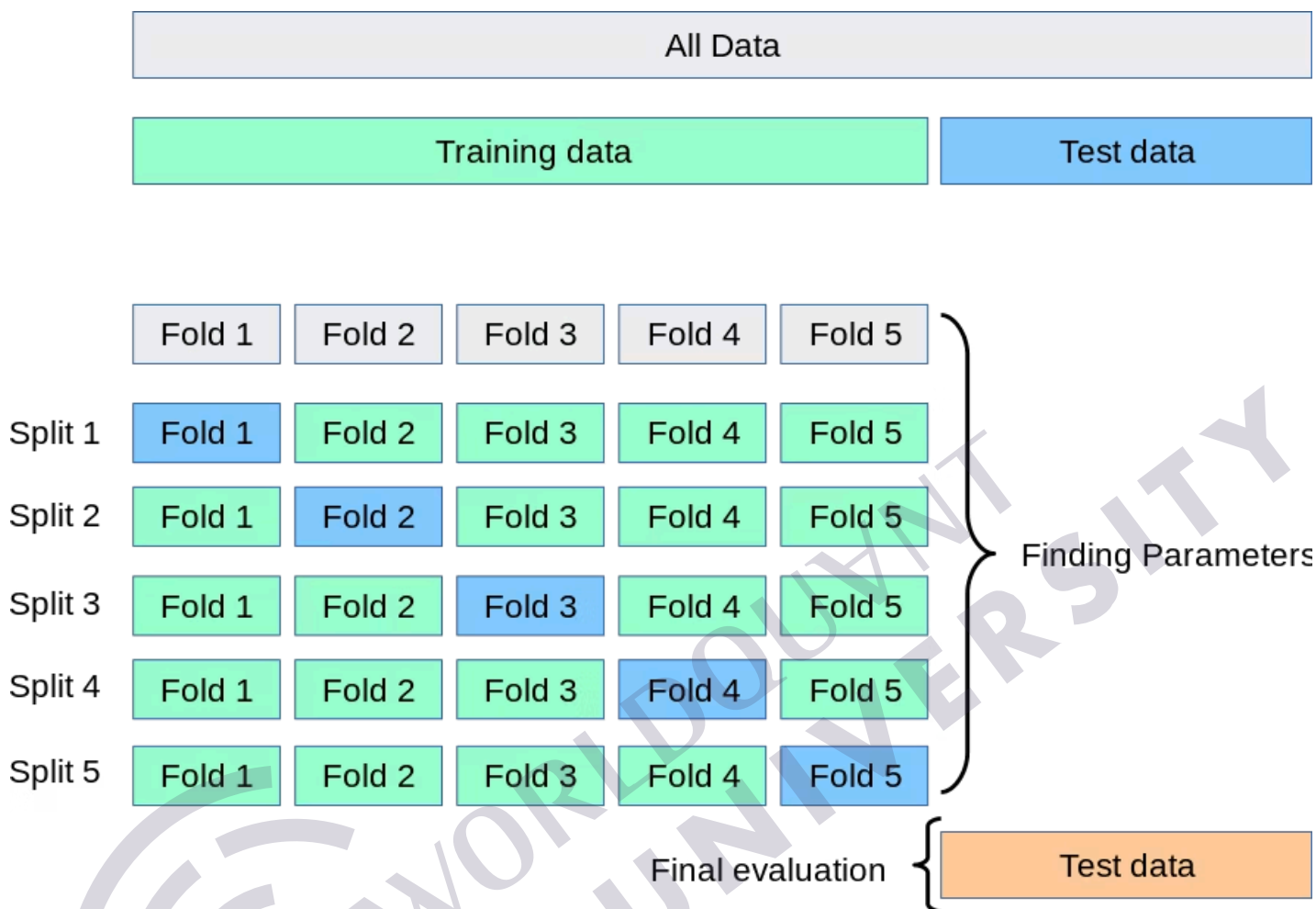


Source: Lv, Dongdong, et al. “Selection of the Optimal Trading Model for Stock Investment in Different Industries.” *PLOS ONE*, 13 Feb 2019. <https://doi.org/10.1371/journal.pone.0212137>.

Cross-validation might be considered an umbrella term for any kind of out-of-sample validation, even including the Walk Forward approach described above. Here, however, we will reserve the term for when the sequential nature of historical data is not necessarily accounted for in the partitioning between training and testing sets. The most basic form of cross-validation is k-fold cross-validation,

in which you split your data into k smaller sets, called folds. The model is then trained using $k-1$ folds and then tested on the remaining fold. These two steps can be repeated for a total of k iterations. The test result is the average of the model's performance over the (up to k) test folds. See the diagram below for a visualization.

Figure 2: K-fold Cross-Validation ($k=5$)



Source: scikit-learn developers_. "3.1 Cross-validation: Evaluating Estimator Performance." scikit-learn_, 2011. https://scikit-learn.org/stable/modules/cross_validation.html.

As you can see, k -fold cross-validation does not necessarily respect the sequential nature of time-series data. As a consequence, it will show poor results for a model or strategy that relies heavily on the assumption that a time series is autocorrelated and/or non-stationary. For example, look again at the diagram above. Let's assume that volatility was very high at the beginning of this historical data set and it gradually decreased; that is, volatility is very high in Fold 1 and gradually decreases so that Fold 5 represents a very "low-vol" environment. Now assume that your strategy incorporates a GARCH (Generalized Autoregressive Conditional Heteroscedasticity) model for volatility: Since the volatility was low in Fold 5, then the strategy has been optimized for low volatility when it is tested in "Split 1" using Fold 1 data. This may not seem *fair* to the autoregressive model. Indeed, k -fold cross-validation and walk-forward backtesting both have advantages and disadvantages, which we will discuss in more detail in the next lesson. For now, let us resolve this apparent flaw in cross-validation for time series by acknowledging that it may not work well for the parameters to be optimized, but can still provide valuable insights into the hyperparameters such as the choice factors included in and excluded from a factor model. On the other hand, walk forward backtesting is more straightforwardly appropriate for both optimized parameters as well as hyperparameters.

1.2 What Backtesting is Not

As we can see from the short review of the two backtesting approaches above, backtesting is by necessity backward-looking (which is why they call it `_back_testing`): it only tells us how the strategy might have performed; it does not provide direct information about how the strategy will perform because the past does not repeat itself. In other words, finance is not a physical science where we might have the benefit of a lab to conduct controlled experiments (de Prado, "Detection" 3). In other words, "As a result, backtests cannot help us derive the precise cause-effect mechanisms that make a strategy successful" (de Prado, *Asset Managers* 105). And for this reason, a backtest "does not prove anything" (de Prado, *Advances* 151).

Given the practical inadequacy of backtesting when it comes to increasing information or knowledge about phenomena, de Prado has a stark warning for all of us as hands-on researchers and hypothesis testers: "Researching and backtesting is like drinking and driving. Do not research under the influence of a backtest." (de Prado, *Advances* 151). Elsewhere, de Prado elaborates, "If the backtest fails to identify a profitable strategy, start from scratch. Resist the temptation of reusing those results" (de Prado, "Slides" 12).

Basically, de Prado is discouraging what might be called the *shotgun* approach to the hunt for a valid strategy. A shotgun requires less accuracy by shooting dozens or hundreds of small pellets that reach a larger circumference; given the dispersion, a target is more likely to be hit—but de Prado might say this is sloppy science. Each experiment or investigation should be guided by the domain knowledge and understanding that comes with subject matter expertise, not random or haphazard trial-and-error informed by guesses and luck.

2. Seven Sins of Backtesting

While we are on the subject of backtesting-related vices, such as researching under the influence of a backtest, let us take heed of the "Seven Sins of Quantitative Investing" published by a team of quantitative analysts at Deutsche Bank in 2014 (de Prado, *Advances* 152). The "seven sins" are listed here (Nousiainen 17):

1. Survivorship bias: Using only companies currently in business ignoring bankruptcy, delisting, and acquisitions.
2. Look-ahead bias: Utilizing non-public information or information not available at the moment of the historical simulation.
3. Storytelling: Creating a story ex-post to justify random patterns.
4. Data mining and data snooping: Training the model on the testing set.
5. Transaction costs: The only way to be sure about the cost is to make the actual trade.
6. Outliers: Building strategy on a few extreme outcomes that may never happen again
7. Shorting: Taking a short position requires finding a lender, and the costs may be unknown.

In the forum, provide an original example of one of these seven sins, explain how it would impair the value of a backtest, and describe how you would avoid or mitigate the "sin."

3. Overcoming SBuMT (Selection Bias under Multiple Testing)

Number four on the list above includes what de Prado calls reusing test results or researching under the influence of a backtest. It skips the vital step of carefully constructing a well-considered hypothesis, validating it, and then testing it independently. This shotgun approach to quantitative finance is so prevalent, that de Prado calls it "[The Multiple Testing Crisis in Financial Research](#)," also the title of the first required reading for this lesson. In it, he generalizes "selection bias under multiple testing" (SBuMT) as to "encompass any situation where we select one outcome, without controlling for the totality of alternative outcomes from which we choose" (de Prado, "Solution" 2). De Prado's point is that it is a distortion of truth and an abuse of science whenever a researcher (or, in other contexts, a pollster or a journalist or anyone else) takes a sample from a population and then generalizes about that population based on, say, the minimum or maximum of that sample—instead of reporting the mean, median, and/or mode; the dispersion (quantiles or standard deviation); and perhaps actual sample size vs. the population size. Consider in this regard what you learned in previous lessons about p-hacking and publication bias in Module 6, Lesson 1. These are just research examples of what we know more broadly as "cherry picking," mentioned as early as the Financial Markets course in relation to the selection of mortgages for securitization: An originator naturally wants to get the less-creditworthy loans off of its book and into the SPV.

Luckily, de Prado has a list of recommendations for overcoming this bias and the resulting crisis. As is so often the case, mere disclosure goes a long way toward solving the problem: Let your audience know how many tests you performed to get your results, and let them draw their own conclusions.

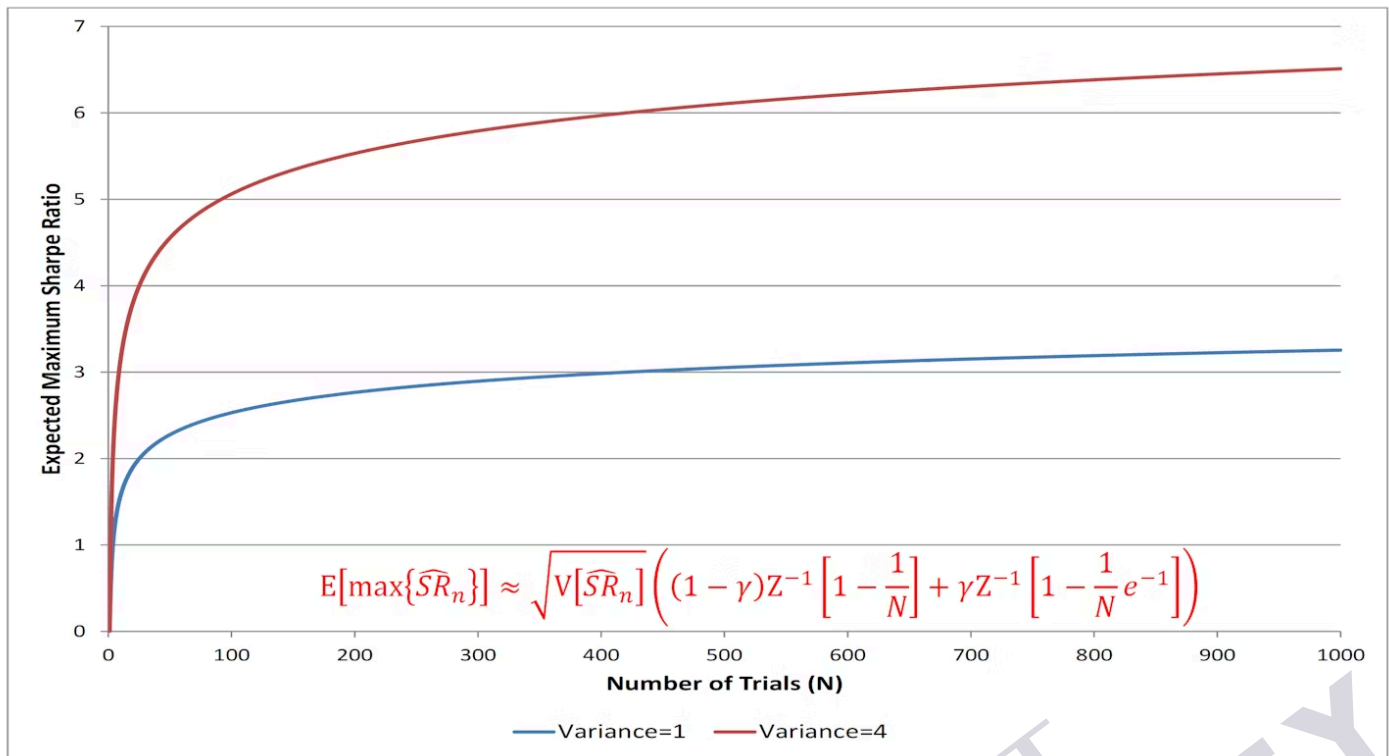
4. Deflated Sharpe Ratio

Perhaps more interesting to us as financial engineers are the statistical approaches he takes to quantify the problem. These involve techniques with which are happily familiar:

- 1) Clustering informed by correlation and quality as we performed in Module 7 of this course. In addition to the metric for cluster quality from Module 7, Lesson 4, you should recognize the Optimal Number of Clusters (ONC) algorithm.
- 2) Recognizing the information value of exceptions to the expected distribution of a random variable. (Think back to the Marchenko–Pastur distribution of eigenvalues from a random matrix in Module 7, Lesson 1.)

Notice in figure 3 how the expected maximum Sharpe ratio increases with the number of trials. "This is a consequence of pure random behavior. We will find high SR strategies "even if there is no investment skill associated with this strategy class" _ _ (de Prado, "Slides" 5). This leads us to the robust deflated Sharpe ratio (DSR), which helps us weed out these worthless SR strategies. Knowing the expected distribution of the Sharpe ratio (SR), "it is possible to derive the probability that the observed SR exceeds a given threshold" (de Prado, "Detection" 4).

Figure 3: Increasing Expected Maximum Sharpe Ratio as a Function of Number of Trials



Source: López de Prado, Marcos. "Advances in Financial Machine Learning: Lecture 5/10 (seminar slides)." 29 Sep 2018, slide 5. <https://ssrn.com/abstract=3257497>.

As before, we have the benefit of a Python repo that implements all of this for us. All we need to do is understand the theory, recognize how the code snippets apply the theory, and play (or, if you prefer, "experiment") with the code to improve our understanding of both: Make sure to read and review this repo as a required reading (Moen "ch8_testing_set_overfitting.py").

5. Conclusion

In this lesson, we took a closer look at backtesting, a concept we had discussed in depth before. We talked about how backtesting should and should not be employed. In addition, we learned some practices and techniques to avoid the deception that can result from improper backtesting.

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